Minimum Targeted Investment to Top a Multi-Task Leaderboard Depends on the Aggregation Rule

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Abstract

Public leaderboards rank models by combining scores on many tasks, and developers can raise a model's scores on chosen tasks. We ask how much targeted improvement a model needs to reach first place and how this depends on the aggregation rule. We formalise a score counterfactual in which one target raises its scores up to a cap while all rivals stay fixed, and report three separate optima (total points, tasks touched, largest single-task lift) and, separately, whether first place is attainable at all. We give closed forms for the mean, median and minimum, and solve the geometric mean exactly and the Borda and Condorcet-winner requirements by integer programming. On the Open LLM Leaderboard (4,495 targets, six tasks) and HELM MMLU (78 targets, 57 subjects), first place is out of reach for most models if no task may be raised by more than 10 points: it is attainable for 3–29% of Open LLM Leaderboard targets and 18–51% of HELM targets, depending on the rule. The rules differ in how many models can win, not only in cost; on the cohort feasible under every rule with loose caps, the minimum rule needs the fewest points and Borda the most tasks. The ordering of rules changes between datasets and caps, so our conclusions are conditional on the cap and the score scale. Investment here is a score counterfactual, not a training cost.

1 Introduction

Public leaderboards turn a score matrix of models by tasks into a single ranking, and the top place carries visibility. The ranking depends on the aggregation rule, and developers can raise a model's score on chosen tasks, by targeted data or tuning. We ask a counterfactual question about the scores: *how much targeted improvement does a given model need to reach first place, and how does that depend on the rule that aggregates the tasks?* We answer it for a single target model with all rivals held fixed, on two public matrices, the Open LLM Leaderboard (OLLB; 4,496 models, six tasks) and HELM MMLU (79 models, 57 subjects).

The question has three parts that earlier work on task selection and aggregation does not separate (Section 2). First, "how much" is not one number: we report the total points added (L_1^*), the number of tasks touched (K^*) and the largest single-task lift (L_∞^*) as three separate optima. Second, an investment is meaningful only under a cap on how far each score can rise, and under attainable caps first place is often out of reach; we therefore report *feasibility* (whether first place is attainable at all, and the smallest cap or uniform lift that makes it so) as the headline, and compare costs only on the cohort of targets that are feasible under every compared rule. Third, the answer may depend on the score scale and the number of tasks, which we vary.

Contributions. (i) We formalise the score-edit counterfactual and its three optima, with feasibility kept separate from cost, for score rules (mean, median, minimum, geometric mean) and ordinal rules

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(Borda, strict Condorcet winner; the latter is an upper bound for any Condorcet-consistent rule). (ii) We give closed forms for the mean, median (odd n) and minimum, and show that L^*_∞ equals the smallest uniform lift for every rule; we check them against exact solvers, including integer programs for the ordinal rules, on 31,080 instances, all of which agree (Section 3, Appendix A). (iii) We run all rules over eight cap scenarios on 4,495 OLLB and 78 HELM targets (Section 4).

Findings. They are conditional on the cap and the scale and are stated as such. **(F1)** If no task may be raised by more than 10 points, first place is attainable for only 3–29% of OLLB targets and 18–51% of HELM targets, depending on the rule; for models ranked 2–30 under the mean the shares are 76–100% and 41–100%. Rules differ in *how many* models can win, not only in what it costs: on OLLB, Borda reaches 29% at 10 points where the mean reaches 3%. **(F2)** The ordering of rules is not stable across datasets and caps: the minimum rule is the hardest to satisfy under small caps on HELM (18% at 10 points) and the easiest on OLLB under a 20-point cap (64% against 14–21% for the mean, median and geometric mean). **(F3)** Under loose caps, on the cohort of targets feasible under every rule, the minimum rule needs the fewest total points on OLLB and, with the median, on HELM, while Borda needs the most tasks (5 of 6 on OLLB, 46 of 57 on HELM). **(F4)** On OLLB, whose six tasks have unrelated scales, the number of tasks needed by targets ranked beyond 10 changes by up to four with the score scale (official, raw, z-score, rank percentile), so OLLB conclusions are tied to the official scale. **(F5)** The closed forms explain these differences: the minimum rule requires every task below the best rival's level to reach it, so it is blocked by any binding cap there.

Investment here is a score counterfactual, not a training cost; it ignores spillover between tasks, rival responses and the data or compute that would deliver the gain (Section 5).

2 Related Work

Aggregation and task selection in multi-task evaluation. Multi-task results depend on which tasks are included and how they are combined. Dehghani et al. [2021] show that rankings can change with task selection and discuss macro-averaging, arithmetic versus geometric means and rank-based aggregation; Liang et al. [2023] leave a universal single-number aggregate explicitly unresolved and stress that it encodes value judgements. These works vary the benchmark while the compared models stay as they are. We instead hold the benchmark and all rivals fixed and edit one model's scores. They do not establish an ordering of rules by the cost of changing a ranking, and we do not read one into them.

Social choice and evaluation. Lanctot et al. [2023] cast evaluation as voting, with models as alternatives and tasks as voters, so that comparisons need only within-task order. The Handbook of Computational Social Choice [Brandt et al., 2016] supplies the voting-rule and Condorcet vocabulary we use [Zwicker, 2016]. Our target-promotion problem is related to priced bribery, where a fixed set of voters is paid to change their votes [Faliszewski and Rothe, 2016]; it is a structured special case in which the target may only move up on the tasks it is allowed to edit, so complexity results for unrestricted bribery do not transfer automatically. We make no claim of new algorithms for bribery or control.

Leaderboard incentives and strategic post-training. Singh et al. [2025] document private testing, selective disclosure and data-access asymmetries on Chatbot Arena, and find in a controlled training-mixture experiment that gains on the targeted evaluation need not transfer broadly. Chen et al. [2026] analyse simultaneous developers' effort and a designer's tuning policy as an equilibrium problem, under assumptions on the score response to effort. Our setting is different and narrower: a single target, fixed rivals, observed score matrices, and no equilibrium, dynamics or latent capability. It measures the *score improvement* that a ranking goal requires under a stated cap, not the training effort that would deliver it.

Benchmark-related training and contamination. Zhou et al. [2023] show that continued training on benchmark-related data raises benchmark scores, with possible losses elsewhere, and Yang et al. [2023] show that rephrased test samples can evade common decontamination while still inflating scores. These works motivate the intervention class we study; they do not provide a task-response or cost law, and they show that independent monotone gains with no side effects are an abstraction

(Section 5). Score tables alone neither identify nor exclude contamination, and nothing here accuses any listed model.

Benchmarks and data used. We use the HELM MMLU release [Liang et al., 2023, Hendrycks et al., 2021] and the Open LLM Leaderboard, whose tasks include BBH [Suzgun et al., 2022] and which is built on the Evaluation Harness [Sutawika et al., 2023].

3 Problem Setting and Method

Leaderboard and rules. A leaderboard is a score matrix $x \in [0, 100]^{M \times n}$ of M models on n tasks (sub-datasets). An aggregation rule R maps a matrix to a ranking of the models. We study score rules, which compute a scalar per model: the arithmetic *mean* (the equal-task mean; on the Open LLM Leaderboard, the mean of the six official normalised task scores is the official average, while the raw-accuracy columns are a separate variant; we do not claim that it reproduces HELM's own aggregate), the *median*, the *minimum*, and the *geometric mean*; and two ordinal rules, which use only within-task order: the *Borda* count and the requirement to be a strict *Condorcet winner* (beating every rival in pairwise comparison, where a task won counts 1, a task tied counts $\frac{1}{2}$, and the target wins if its total exceeds $n/2$; with ties this is not the same as winning more than $n/2$ tasks strictly).

Intervention and investment. Fix a target model i. The developer may raise its scores to any $y_t \in [x_{it}, c_t]$, where c_t is a cap on task t; all rivals' raw scores stay fixed, there are no spillovers to other tasks, and no rival responds. After the edit, ordinal aggregates of rivals are recomputed because the target may pass them. The target is *first* if it beats every rival under R by a margin ε (score rules; $\varepsilon = 10^{-3}$ points unless stated) or strictly (ordinal rules). Strict victory over a continuous score can have an infimum that is not attained, so ε defines the margin version we report, and we report sensitivity to it; a target that already wins has cost zero.

Investment is a score counterfactual, not a training cost. We measure it by three separate optima, each its own optimisation problem: L_1^*, the least total number of points added; K^*, the fewest tasks touched; and L_∞^*, the smallest largest single-task lift when any subset may be raised. Where a solver minimises one of them, the others are only reported descriptors at its solution. Raising all tasks by a common amount is a different quantity and is not called L_∞^*. A target is *infeasible* if no allowed target row under the cap satisfies the winning condition (a cap that binds at the optimum does not make a target infeasible); we report the feasible fraction separately from costs over feasible targets, never averaging an infinite cost, together with the smallest cap at which the target becomes feasible. For the Condorcet requirement, the cost is that of a sufficient condition and therefore an upper bound for any particular Condorcet-consistent rule (Schulze, Kemeny, ...); it is not that rule's exact cost.

Caps, scales and cohort (modelling assumptions). The cap is the central assumption and the headline result is investment as a function of it. We use a loose ceiling ($c_t = 100$), a frontier cap (best observed score on t plus a margin, never below the current score), and an empirical cap (the current score plus a high percentile of observed fine-tune-minus-base score gains on t, never below the current score and never above the ceiling); further caps appear in the cap sweep. Frontier and empirical caps are assumptions, not evidence that such scores are attainable; empirical caps carry survivorship bias and use one gain quantile for all targets. Open LLM Leaderboard scores are on unrelated scales, so raw scores, per-task standardised scores and rank percentiles are distinct variants (Section 4). Geometric means use only positive-domain scales; z-scores are used only for mean, median, minimum and pooled L_1. Transform parameters are fixed from the original snapshot and edits and caps are mapped through the same transform. In every comparison across rules the targets, rivals and caps are identical (a fixed cohort); where the ordinal solvers are too slow, a rank-stratified sample is used for *all* rules in that table and results are reported by rank bin. Task bootstraps and evaluation-noise simulations are sensitivity analyses conditional on these assumptions and do not validate them.

Closed forms for the score rules. Let τ be the best rival's value under the rule, $b = \tau + \varepsilon$, $h_t = \max(c_t - x_{it}, 0)$ the headroom, and $x = x_{i\cdot}$. Write d^* for the smallest d such that $y = \min(x + d, c)$ makes the target first: since every rule is monotone in the target's own scores, any set of edits that wins can be replaced by this uniform-lift edit with no larger largest lift, so $L_\infty^* = d^*$ for every rule (Proposition 4).

1. **Mean.** Let $\text{need} = nb - \sum_t x_t$. The target is already first if $\text{need} \leq 0$; it is feasible iff $\sum_t h_t \geq \text{need}$; then $L_1^* = \text{need}$ and K^* is the fewest tasks whose largest headrooms sum to at least need.
2. **Median, n odd.** The target needs at least $m = (n+1)/2$ tasks with $y_t \geq b$. With $a = \#\{t : x_t \geq b\}$, $K^* = \max(0, m - a)$; L_1^* is the sum of the K^* smallest values of $b - x_t$ over tasks below b with $c_t \geq b$; it is infeasible iff fewer than K^* such tasks exist. (For even n, as on the six-task Open LLM Leaderboard, the median is the mean of two middle values and the solver is used; it is checked against brute force.)
3. **Minimum.** $K^* = \#\{t : x_t < b\}$, $L_1^* = \sum_t (b - x_t)^+$, feasible iff $c_t \geq b$ on every task below b.

These formulas explain the cap dependence we find: a rule that requires every task to reach b (minimum) or a majority of tasks (median) is blocked as soon as the cap lies below b on those tasks, whereas the mean can trade headroom across tasks. They are conditional on the model above; they are not statements about training. We verify the formulas, and Proposition 4 for the geometric mean, Borda and Condorcet as well, against the exact solvers on task subsets, targets and caps sampled from the two matrices (31,080 checks, all agreeing; Appendix, Table 3); the proofs are short and are given there.

4 Experiments

Data and cohorts. OLLB: the `contents` table of the Open LLM Leaderboard v2 (pinned commit `9c09a7ca`), with its six official normalised task scores (IFEval, BBH, MATH Lvl 5, GPQA, MuSR, MMLU-Pro), 4,496 models after removing 79 duplicate names and one flagged model (submissions 2024-06 to 2025-03). HELM MMLU: release v1.13.0, 79 models, exact-match accuracy on 57 subjects. Both are frozen snapshots; all ranks and costs refer to them. A *target* is any model of rank ≥ 2 under the mean rule: 4,495 on OLLB and 78 on HELM. Score rules run on every target. The ordinal solvers are slower, so Borda and Condorcet run on a rank-stratified sample of 136 OLLB targets (ranks 2–21 and log-uniform tail draws) with stratum weights that make weighted shares and medians estimate the full-leaderboard quantity; HELM runs every rule on every target. We report by rank group of the target under the mean rule (2–10, 11–100, >100). Caps (Section 3): ceiling (100), frontier + 2 (best rival + 2 points), $+5/10/20$ points over the target's own score, and the empirical 95th and 99th percentiles of fine-tune-minus-base gains (OLLB, 540 base/fine-tune pairs). For HELM there are no such pairs, so empirical caps use the OLLB MMLU-Pro gain percentile for every subject, a proxy marked † throughout. Intervals are 95% percentile bootstraps over targets within rank strata; they describe these snapshots, not a population of leaderboards. OLLB values are normalised points and HELM values percentage points, so costs are compared across rules, never across datasets. The solvers are exact (closed forms, convex or greedy solvers, and HiGHS integer programs), and every solver status is either recorded as optimal or cross-checked against an independent feasibility test (Appendix D).

Headline: how many targets can reach first place at all. Figure 1 shows the share of targets for which first place is attainable when no task may be raised by more than h points, i.e. the distribution of the smallest uniform lift d^* (a target is feasible under a gain cap h exactly when $d^* \leq h$, which we verified on every row). Table 1 adds the empirical cap and the rank groups. At a 10-point cap the shares of all OLLB targets are 3% (mean), 5% (median), 10% (minimum), 3% (geometric mean), 29% (Borda) and 3% (Condorcet winner); on HELM they are 37%, 51%, 18%, 31%, 31% and 49%. Almost all models far from the top cannot reach first place: of the OLLB targets ranked beyond 100 at most 28% (Borda; 9% for the minimum) and under the empirical 95% cap at most 25% can (Table 1). For the 29 best-placed targets (ranks 2–30) the shares are high, 76–100% on OLLB and 41–100% on HELM, and the minimum rule is lowest on both. The rules change *who* can win, not only the price: at a 20-point cap the minimum, Borda and Condorcet rules are attainable for 46–64% of OLLB targets against 14–21% for the mean, median and geometric mean, whereas on HELM the minimum rule is the hardest at 5, 10 and 20 points (9%, 18% and 44%). The empirical cap is stricter: under the 95th-percentile cap only the targets near the top qualify (OLLB ranks 2–10: 100% for mean, median, geometric mean and Borda, 44–56% for minimum and Condorcet; ranks 11–100: 22–100%), and on HELM, with the proxy cap, at most 1% of targets of rank 11–100 for any rule. The smallest uniform

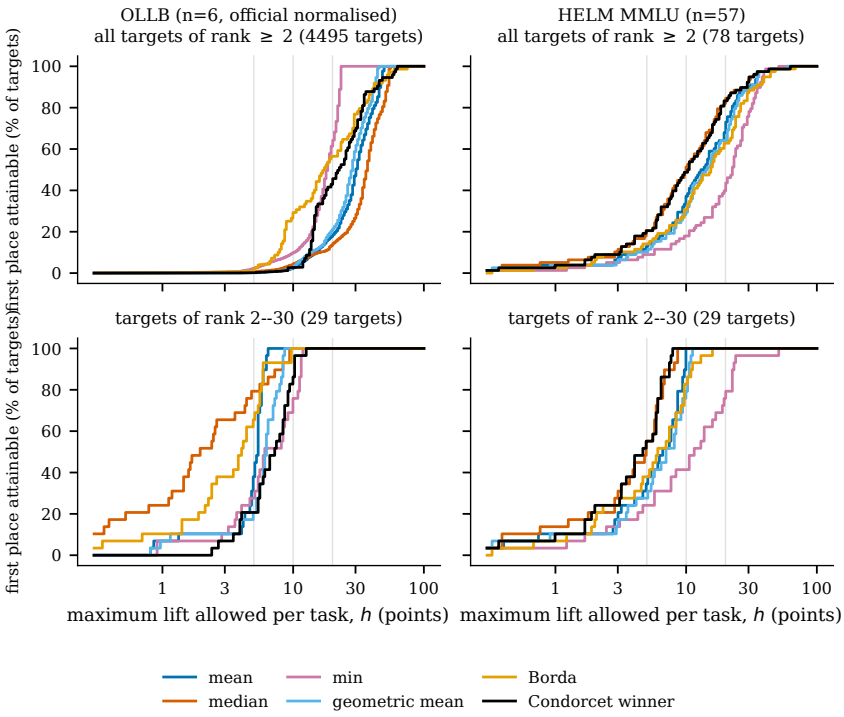


Figure 1: Share of targets for which first place is attainable when no task may be raised by more than h points (weighted empirical CDF of the minimal uniform lift d^*; vertical lines at $h = 5, 10, 20$). Top: all targets of rank ≥ 2 (OLLB 4,495; HELM 78); bottom: the 29 targets ranked 2–30. Borda and Condorcet on OLLB: 136-target stratified sample, stratum-weighted.

Table 1: Feasibility of first place by rule and the target's rank under the mean (2–10, 11–100, >100): share of targets for which first place is attainable with at most 10 points per task, and under the empirical 95th-percentile gain cap; and the minimal uniform lift d^* in points, median [95% bootstrap interval, omitted where narrower than 1 point]. OLLB: 4,495 targets (score rules), 136-target stratified sample (Borda, Condorcet); HELM: 78 targets, none ranked beyond 79. †HELM empirical caps are a proxy.

	feasible, cap +10			feasible, emp. 95%			min. lift d^* (pts)		
rule	2–10	11–100	>100	2–10	11–100	>100	2–10	11–100	>100
OLLB									
mean	100%	100%	0%	100%	100%	5%	4	8	30
median	100%	73%	3%	100%	61%	2%	0 [0,3]	7 [7,8]	36
min	78%	68%	9%	44%	23%	1%	8 [5,11]	8 [6,9]	18
geo. mean	100%	81%	1%	100%	47%	1%	6 [5,6]	8 [8,9]	28
Borda	100%	96%	28%	100%	82%	25%	2 [2,5]	5 [4,6]	18 [15,23]
Condorcet	100%	58%	1%	56%	22%	0%	5	9 [8,11]	23 [18,26]
HELM: emp. 95% caps are a proxy† (OLLB MMLU-PRO gains)									
mean	100%	29%	–	78%	0%	–	3	15 [12,19]	–
median	100%	45%	–	100%	1%	–	1 [0,3]	11 [10,14]	–
min	67%	12%	–	56%	0%	–	3 [2,11]	23 [21,26]	–
geo. mean	100%	22%	–	56%	0%	–	3	15 [13,20]	–
Borda	100%	23%	–	89%	0%	–	2	16 [15,21]	–
Condorcet	100%	42%	–	100%	1%	–	2	11 [10,14]	–

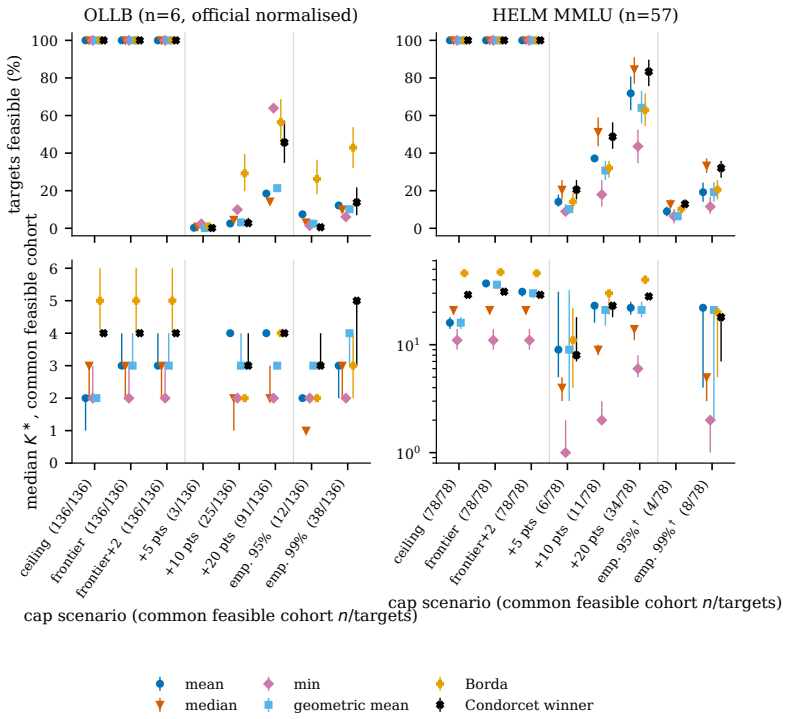


Figure 2: Cap sweep. Top: share of targets (rank ≥ 2) for which first place is attainable, by rule and cap scenario (caps are not nested; † proxy). Bottom: median K^* on the common feasible cohort of that cap (tick label: cohort size / targets run); points are shown for cohorts of at least 5 targets, bars: 95% bootstrap intervals. Cohorts differ between caps: compare rules within one cap.

lift grows with rank under every rule (OLLB median d^* of 0–8 points for ranks 2–10, 5–9 for ranks 11–100, 18–36 beyond rank 100; HELM 11–23 points for ranks 11–100).

Cost on the common feasible cohort. Costs are compared only among targets feasible under every listed rule at that cap (Table 2). At the ceiling cap on OLLB the minimum rule needs the fewest points (25.3 against 100.6 for the mean), then the median (48.1), Borda (65.6), Condorcet (78.1) and the geometric mean (78.8). On HELM the minimum (98) and median (111) rules need the fewest points and the mean (661) and geometric mean (633) the most, with Condorcet (213) and Borda (500) in between; relative to the mean's L_1^*, the median rule's median is 0.15 (95% interval 0.14–0.19) and the minimum's 0.14 on HELM, and 0.60 and 0.25 on OLLB (`rel_L1_vs_mean`). The number of tasks tells a different story: Borda needs the most tasks (OLLB 5 of 6; HELM 46 of 57), the minimum rule the fewest on HELM (11), and the mean rule's K^* depends strongly on the cap, because a binding cap forces the total to be spread over more tasks (HELM: 16 at the ceiling, 31 at frontier + 2). So the rule that is cheapest in points is not the one that touches the fewest tasks, and the ranking of rules depends on which investment measure is used; on both datasets the ordinal rules need many tasks but not many points relative to the mean.

Sensitivity to the cap, rank, number of tasks and scale. Figure 2 sweeps all cap scenarios. Ceiling and frontier caps leave every target feasible, while at $+5$ points 0–2% of OLLB targets are feasible and at $+20$ points 14–64%, so the feasibility ordering of the rules shown above is what changes with the cap; we do not rank costs across rules at attainable caps, where cohorts are small (3 to 91 targets) and differ. Cost rises with the target's rank for every rule overall, though not strictly at every rank (Appendix Figure 3). On HELM random subsets of n subjects (100 draws per n) show that Borda and Condorcet need more tasks as n grows, so part of the Borda cost above reflects $n = 57$ (Appendix Figure 4). On OLLB, replacing the official normalised scores with raw accuracies, per-task z-scores or rank percentiles changes median K^* for targets ranked 3–10 by at most one task, but for targets ranked beyond 10 by up to four: at the ceiling the mean rule needs 2 tasks on the official scale and 6 on rank percentiles, and the minimum rule 3 against 5 (raw) and 6 (z-score)

Table 2: Cost of first place on the common feasible cohort at a cap, median [95% bootstrap interval] of tasks touched K^* and total points L_1^*. The cohort and the targets run are in each block header. OLLB values are normalised points, HELM values percentage points, so they are not comparable across datasets; cohorts differ between blocks, so compare rules within a block. The ceiling and frontier + 2 caps leave every target feasible; cohorts at attainable caps are small and are in Appendix B.

rule	K^*	L_1^* (points)
OLLB, cap ceiling (100): common feasible cohort 136 of 136 targets		
mean	2 [1,2]	100.6 [83.7,129.7]
median	3 [2,3]	48.1 [37.0,66.3]
min	2 [2,3]	25.3 [22.8,27.5]
geometric mean	2 [2,2]	78.8 [58.5,101.7]
Borda	5 [4,6]	65.6 [48.8,94.5]
Condorcet winner	4 [4,4]	78.1 [56.6,97.6]
OLLB, cap frontier+2: common feasible cohort 136 of 136 targets		
mean	3 [3,4]	100.6 [83.7,129.7]
median	3 [2,3]	48.1 [37.0,66.3]
min	2 [2,3]	25.3 [22.8,27.5]
geometric mean	3 [3,4]	81.9 [63.7,107.5]
Borda	5 [4,6]	65.6 [48.8,94.5]
Condorcet winner	4 [4,4]	78.1 [56.6,97.6]
HELM, cap ceiling (100): common feasible cohort 78 of 78 targets		
mean	16 [14,18]	661 [601,787]
median	21 [19,22]	111 [87,135]
min	11 [9,14]	98 [82,127]
geometric mean	16 [14,18]	633 [575,758]
Borda	46 [45,49]	500 [438,621]
Condorcet winner	29 [29,29]	213 [174,228]
HELM, cap frontier+2: common feasible cohort 78 of 78 targets		
mean	31 [28,32]	661 [601,787]
median	21 [19,22]	111 [87,135]
min	11 [9,14]	98 [82,127]
geometric mean	30 [27,31]	641 [582,767]
Borda	46 [45,49]	500 [438,621]
Condorcet winner	29 [29,29]	213 [174,228]

(Appendix Figure 5). We therefore state OLLB conclusions for the official scale and treat the others as sensitivity.

Robustness and theory check. Changing the strictness margin ε from 10^{-3} to 10^{-2} or 10^{-1} changes the median L_1^* by at most 12%; K^* changes for at most 13% of targets at the ceiling cap (Table 4). Removing merged models leaves median K^* at the ceiling and frontier caps unchanged; keeping only the 500 top-ranked models leaves it unchanged for targets ranked 3–10 and lowers it by one to two tasks for lower-ranked targets (Table 6). An evaluation-noise Monte Carlo and a task bootstrap on 10 targets (100–200 draws) move the feasible share of the minimum rule from the observed 70% to 49–70% and leave median K^* within one task of the observed value on OLLB and within up to eight tasks (Borda) on HELM (Table 7); these are conditional sensitivity analyses on few targets, not estimates of leaderboard sampling variation. The closed forms (mean, median for odd n, minimum) and the uniform-lift identity agree with the solvers on all 31,080 instances (Table 3).

5 Discussion and Limitations

What the results say. Under attainable caps, most models cannot reach first place under any of the six rules, and the rule determines how many can: the share at 10 points ranges from 3% to 29% on OLLB and from 18% to 51% on HELM. Cost comparisons across rules are meaningful only where the same targets are feasible under all of them; there, the minimum and median rules need the fewest points, Borda the most tasks, and the mean rule's task count depends on the cap. The ordering of rules differs between OLLB and HELM and between caps (Figures 1 and 2), so the supported statement is conditional: it holds per cap scenario and scale, not as a single ranking of rules. The closed forms

identify why: the minimum rule needs every task below the best rival's level to reach it, the median rule a majority, and the mean rule only a total, which any cap that does not bind can supply from few tasks.

Limitations. *The estimand is a score counterfactual, not a training cost.* It assumes that any score in $[x_{it}, c_t]$ is reachable independently on each task, with no spillover (gains on one task need not transfer, and may reduce other scores), no rival response and no equilibrium, and it says nothing about the data, compute or contamination that would deliver the gain. *Caps are assumptions.* Frontier and empirical caps are not evidence that such scores are attainable; the empirical caps come from observed fine-tune-minus-base pairs (survivorship bias, one quantile for all targets), and on HELM they are a proxy borrowed from another benchmark. The conclusions on feasibility are therefore as good as the cap scenario; we show eight and rank nothing across them. *Scale and size.* OLLB has $n = 6$ tasks on unrelated scales, so median, minimum and geometric-mean results are tied to the official normalisation (Appendix Figure 5) and the median and Condorcet results are tie-sensitive; HELM is one benchmark with 57 same-unit subjects and 79 models, and describes a snapshot, not a population of leaderboards. *Cohorts and intervals.* Borda and Condorcet on OLLB are computed on a 136-target stratified sample (stratum-weighted); cohorts at attainable caps are small (3 to 91 targets); the Condorcet cost is an upper bound for any Condorcet-consistent rule; K^* has few levels at $n = 6$. The task bootstrap and the evaluation-noise simulations use 10 targets and are sensitivity analyses conditional on the model and caps; they do not validate them. *Margin.* Strict victory over a continuous score has an infimum, so we fix $\varepsilon = 10^{-3}$ and report sensitivity to it (Table 4). *Measurement and data.* Scores are single-run leaderboard numbers; one OLLB model per name is kept; the licences of the score data are not declared on the sources we could verify. We use no evidence about any model's training data, and nothing here accuses any listed model.

Broader impact. The study documents how little targeted improvement some rules require and how others are out of reach, which can inform more robust aggregation and also show where leaderboards are easiest to game. It uses public score tables only, with no human subjects and no model training.

Conclusion. First place under a multi-task rule is not a single price: it is a feasibility question first and a cost question only among targets for which first place is attainable. On the two matrices studied, the aggregation rule changes how many models can win and what it costs, and neither ordering is stable across datasets and caps.

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A Proofs of the Propositions

Notation as in Section 3: $x = x_{i\cdot}$ is the target's score vector, $c_t \geq x_t$ the cap, $h_t = c_t - x_t \geq 0$ the headroom, τ the best rival's value under the score rule, $b = \tau + \varepsilon$; the target is first iff its value is at least b. Let y be an edit with $x \leq y \leq c$.

Proposition 1 (mean). The target is first iff $\sum_t y_t \geq nb$. With $\mathrm{need} = nb - \sum_t x_t$: if $\mathrm{need} \leq 0$ no edit is needed. Otherwise $\sum_t (y_t - x_t) \geq \mathrm{need}$ is necessary, so $L_1^* \geq \mathrm{need}$; it is attainable iff $\sum_t h_t \geq \mathrm{need}$ (raise any tasks within their headroom until the total is met), which is also the feasibility condition. K^* is the least K such that the K largest headrooms sum to at least need, since an edit touching K tasks adds at most the sum of their headrooms. □

Proposition 2 (median, n odd). The median is at least b iff at least $m = (n+1)/2$ entries are $\geq b$. Let $a = \#\{t : x_t \geq b\}$; those entries already count. Every other counted task must be raised to at least b and so must have $c_t \geq b$ and cost at least $b - x_t$. Hence at least $\max(0, m - a)$ tasks must be touched, $K^* = \max(0, m - a)$ if at least that many tasks below b have $c_t \geq b$, and the target is infeasible otherwise. Raising a chosen task to exactly b suffices, so L_1^* is the sum of the K^* smallest values of $b - x_t$ among those tasks. □

Proposition 3 (minimum). The minimum is at least b iff every $y_t \geq b$. Every task with $x_t < b$ must be raised, to at least b, which needs $c_t \geq b$; raising each to exactly b suffices. So $K^* = \#\{t : x_t < b\}$, $L_1^* = \sum_t (b - x_t)^+$, and the target is feasible iff $c_t \geq b$ on every such task. □

Proposition 4 (L_∞^* equals the smallest uniform lift). Each rule we study is monotone in the following sense: if y makes the target first, so does every y' with $y \leq y' \leq c$. For the score rules this holds because they are nondecreasing in each coordinate and the rivals are fixed. For Borda and the Condorcet requirement, raising the target's score on a task can only improve its position against each rival on that task and cannot improve any rival's, so the target's total never falls and each rival's total never rises. Let y win with largest lift $\delta = \max_t (y_t - x_t)$ and put $y'_t = \min(x_t + \delta, c_t)$. Then $y \leq y' \leq c$, so y' wins, and its largest lift is at most δ. Hence

$L^*_\infty = d^* := \min\{d : \min(x + d, c) \text{ wins}\}$ for every such rule. □ *Corollary.* For the gain cap $c_t = \min(x_t + h, 100)$, a target is feasible iff $d^* \le h$ (with d^* computed at the ceiling), because by monotonicity feasibility is equivalent to winning at $y = c$. The numerical agreement of Propositions 1–4 with the solvers is in Table 3.

B Additional Tables

Table 3: Closed forms (P1 mean, P2 median, P3 minimum; P4: L^*_∞ equals the smallest uniform lift) against the solvers on 31,080 instances sampled from the two matrices (task subsets, targets, caps): all agree; largest absolute differences are at most 10^{-6} (tolerance 10^{-4}); feasibility status agrees on every instance.

prop.	rule	inst.	agree	cap binds	first	max $\lvert\Delta L_1\rvert$	max $\lvert\Delta K\rvert$	max $\lvert\Delta L_\infty\rvert$
P1	mean	2088	2088	849	12	0	0	1e−7
P2	median	2040	2040	929	32	0	0	1e−7
P3	min	2088	2088	1211	20	0	0	9e−8
P4	Borda	6216	6216	2493	0	–	–	1e−6
P4	Condorcet	6216	6216	2337	0	–	–	1e−6
P4	geo. mean	6216	6216	2580	0	–	–	9e−8
P4	mean	2088	2088	849	0	–	–	9e−8
P4	median	2040	2040	929	0	–	–	9e−8
P4	min	2088	2088	1211	0	–	–	9e−8

Table 4: Sensitivity to the strictness margin ε (base 10^{-3}): median ratio of L^*_1 for $\varepsilon = 0.01$ and 0.1 to the base, and the share of targets whose K^* changes. "feas." is the feasible share at the base margin (OLLB Borda and Condorcet: stratified sample). † proxy cap.

data	cap	rule	feas.	ratio 0.01	ratio 0.1	K chg. 0.01	K chg. 0.1
HELM	ceil. (100)	Borda	100%	1.001	1.010	0%	0%
HELM	ceil. (100)	Condorcet	100%	1.001	1.014	0%	0%
HELM	ceil. (100)	geo. mean	100%	1.001	1.008	3%	12%
HELM	ceil. (100)	mean	100%	1.001	1.008	1%	13%
HELM	ceil. (100)	median	100%	1.002	1.019	0%	10%
HELM	ceil. (100)	min	100%	1.001	1.011	1%	9%
HELM	emp. 95%†	Borda	10%	1.009	1.095	0%	38%
HELM	emp. 95%†	Condorcet	13%	1.010	1.114	0%	10%
HELM	emp. 95%†	geo. mean	6%	1.008	1.084	0%	100%
HELM	emp. 95%†	mean	9%	1.003	1.086	0%	100%
HELM	emp. 95%†	median	13%	1.009	1.122	0%	22%
HELM	emp. 95%†	min	6%	1.006	1.064	0%	0%
OLLB	ceil. (100)	Borda	100%	1.002	1.007	0%	0%
OLLB	ceil. (100)	Condorcet	100%	1.000	1.000	0%	0%
OLLB	ceil. (100)	geo. mean	100%	1.000	1.004	0%	1%
OLLB	ceil. (100)	mean	100%	1.000	1.003	0%	1%
OLLB	ceil. (100)	median	100%	1.000	1.003	0%	0%
OLLB	ceil. (100)	min	100%	1.001	1.008	0%	1%
OLLB	emp. 95%	Borda	89%	1.003	1.009	3%	3%
OLLB	emp. 95%	Condorcet	42%	1.000	1.000	0%	0%
OLLB	emp. 95%	geo. mean	2%	1.002	1.018	0%	9%
OLLB	emp. 95%	mean	7%	1.001	1.009	1%	9%
OLLB	emp. 95%	median	3%	1.002	1.021	0%	4%
OLLB	emp. 95%	min	1%	1.003	1.031	0%	2%

C Additional Figures

D Reproducibility, Data and Use of LLMs

Reproducibility. The supplementary code package contains the solvers, the data build with SHA-256 checksums, all result files, the unit tests (solvers against brute force) and a reproduction guide with a script-to-table map. Every number in this paper is read from its result files;

Table 5: Cost of first place on the common feasible cohort at attainable caps (+10 points; empirical 95%†), median [95% bootstrap interval]. Cohorts are 4–25 targets; blocks with fewer than 10 targets are not summarised. † proxy cap on HELM.

rule	K^*	L_1^* (points)
OLLB, cap +10 pts: common feasible cohort 25 of 136 targets		
mean	4 [4,4]	33.3 [31.9,37.1]
median	2 [1,2]	9.6 [4.8,15.4]
min	2 [2,2]	10.8 [6.2,11.6]
geometric mean	3 [3,4]	26.7 [23.7,32.0]
Borda	2 [2,2]	7.5 [6.7,9.1]
Condorcet winner	3 [3,4]	17.7 [15.4,20.7]
OLLB, cap emp. 95%: common feasible cohort 12 of 136 targets		
mean	2 [2,2]	37.5 [33.5,37.5]
median	1 [1,1]	9.6 [9.6,9.6]
min	2 [2,2]	5.9 [5.4,8.5]
geometric mean	3 [3,3]	35.1 [29.9,35.1]
Borda	2 [2,2]	9.9 [9.1,9.9]
Condorcet winner	3 [3,4]	20.4 [16.6,20.4]
HELM, cap +10 pts: common feasible cohort 11 of 78 targets		
mean	23 [16,23]	222 [156,222]
median	9 [8,10]	17 [13,17]
min	2 [2,3]	6 [2,8]
geometric mean	21 [15,21]	209 [142,209]
Borda	30 [19,30]	100 [56,100]
Condorcet winner	23 [18,23]	55 [29,62]
HELM, cap emp. 95%†: common feasible cohort 4 of 78 targets		
mean	*fewer than 10 targets*	
median	*fewer than 10 targets*	
min	*fewer than 10 targets*	
geometric mean	*fewer than 10 targets*	
Borda	*fewer than 10 targets*	
Condorcet winner	*fewer than 10 targets*	

none is typed by hand. Environment: Python 3.11.2, numpy 2.4.6, scipy 1.17.1, pandas 3.0.6, HiGHS through `scipy.optimize.milp`; seed 0 throughout (target sampling, bootstrap, noise draws, theory instances); CPU only. Data: OLLB `contents` at commit `9c09a7ca`…(and `results` at `aa81ecc3`…for the evaluation-noise model) and HELM MMLU release v1.13.0, both public and pinned. Compute: wall-clock on a shared host, 2-CPU jobs under 1 GB each: HELM, all rules and caps, 862 s; OLLB score rules on all targets, 2,164 s; ordinal integer programs on the 136-target sample, 1,698 s (Borda) and 890 s (Condorcet) at the frontier + 2 cap and 1,824 s (Borda) at the ceiling; at $M = 4{,}496$ a Borda solve takes a median of 8.3 s (maximum 115 s). The ordinal runs at frontier + 2 record their solver status (all optimal); older ordinal runs did not, and for them an independent feasibility test agrees with the solver's finite/infinite answer on all of 564,000 checked rows (one early disagreement was re-solved), so their budgets are "solver-reported minimum, gap not recorded". All 350 data-set/cap/rule cells are complete.

Licences and assets. The OLLB `contents` dataset carries no licence declaration on its card that we could verify, and HELM outputs are published by Stanford CRFM in a public bucket without a per-file licence that we checked; neither is redistributed beyond derived score matrices (model names and scores). OLLB's per-model licence column is blank for 1,736 of 4,497 models (before removal of the one flagged model) and is not a licence for the score data. The derived score matrices are part of the code package.

Use of LLMs. All code, experiments, analysis and text, including this manuscript, were produced by LLM agents (Claude, through Claude Code) acting as a research organisation; humans did not edit them. We did not search for or read any publication on this exact question.

Table 6: Leaderboard slices (OLLB score rules): all models, merged models removed, top-500 models only; ranks 3–10 hold 8 targets in every slice. At +10 points, medians are among feasible targets only (feasible share beside them) and are not ranked across rules.

cap	rule	rank	slice	targets	feas.	K^*
ceil. (100)	mean	3-10	all	8	100%	1 [1,1]
ceil. (100)	mean	>10	all	4486	100%	2 [2,2]
ceil. (100)	median	3-10	all	8	100%	1 [1,1]
ceil. (100)	median	>10	all	4486	100%	3 [3,3]
ceil. (100)	min	3-10	all	8	100%	2 [2,2]
ceil. (100)	min	>10	all	4486	100%	3 [3,3]
ceil. (100)	geo. mean	3-10	all	8	100%	1 [1,1]
ceil. (100)	geo. mean	>10	all	4486	100%	3 [3,3]
frontier	mean	3-10	all	8	100%	1 [1,2]
frontier	mean	>10	all	4486	100%	4 [4,4]
frontier	median	3-10	all	8	100%	1 [1,1]
frontier	median	>10	all	4486	100%	3 [3,3]
frontier	min	3-10	all	8	100%	2 [2,2]
frontier	min	>10	all	4486	100%	3 [3,3]
frontier	geo. mean	3-10	all	8	100%	1 [1,2]
frontier	geo. mean	>10	all	4486	100%	5 [5,5]
+10 pts	mean	3-10	all	8	100%	3 [3,3]
+10 pts	mean	>10	all	4486	2%	6 [6,6]
+10 pts	median	3-10	all	8	100%	1 [1,1]
+10 pts	median	>10	all	4486	4%	3 [3,3]
+10 pts	min	3-10	all	8	75%	2 [2,2]
+10 pts	min	>10	all	4486	10%	2 [2,2]
+10 pts	geo. mean	3-10	all	8	100%	3 [2,3]
+10 pts	geo. mean	>10	all	4486	3%	5 [5,6]
ceil. (100)	mean	3-10	no merges	8	100%	1 [1,1]
ceil. (100)	mean	>10	no merges	3779	100%	2 [2,2]
ceil. (100)	median	3-10	no merges	8	100%	1 [1,1]
ceil. (100)	median	>10	no merges	3779	100%	3 [3,3]
ceil. (100)	min	3-10	no merges	8	100%	2 [2,2]
ceil. (100)	min	>10	no merges	3779	100%	3 [3,3]
ceil. (100)	geo. mean	3-10	no merges	8	100%	1 [1,1]
ceil. (100)	geo. mean	>10	no merges	3779	100%	3 [3,3]
frontier	mean	3-10	no merges	8	100%	1 [1,2]
frontier	mean	>10	no merges	3779	100%	4 [4,4]
frontier	median	3-10	no merges	8	100%	1 [1,1]
frontier	median	>10	no merges	3779	100%	3 [3,3]
frontier	min	3-10	no merges	8	100%	2 [2,2]
frontier	min	>10	no merges	3779	100%	3 [3,3]
frontier	geo. mean	3-10	no merges	8	100%	1 [1,2]
frontier	geo. mean	>10	no merges	3779	100%	5 [5,5]
+10 pts	mean	3-10	no merges	8	100%	3 [3,3]
+10 pts	mean	>10	no merges	3779	2%	6 [5,6]
+10 pts	median	3-10	no merges	8	100%	1 [1,1]
+10 pts	median	>10	no merges	3779	4%	3 [2,3]
+10 pts	min	3-10	no merges	8	88%	2 [2,2]
+10 pts	min	>10	no merges	3779	10%	2 [2,2]
+10 pts	geo. mean	3-10	no merges	8	100%	3 [2,3]
+10 pts	geo. mean	>10	no merges	3779	3%	5 [5,6]
ceil. (100)	mean	3-10	top 500	8	100%	1 [1,1]
ceil. (100)	mean	>10	top 500	490	100%	1 [1,1]
ceil. (100)	median	3-10	top 500	8	100%	1 [1,1]
ceil. (100)	median	>10	top 500	490	100%	2 [2,2]
ceil. (100)	min	3-10	top 500	8	100%	2 [2,2]
ceil. (100)	min	>10	top 500	490	100%	2 [2,2]
ceil. (100)	geo. mean	3-10	top 500	8	100%	1 [1,1]
ceil. (100)	geo. mean	>10	top 500	490	100%	1 [1,1]
frontier	mean	3-10	top 500	8	100%	1 [1,2]
frontier	mean	>10	top 500	490	100%	3 [3,3]
frontier	median	3-10	top 500	8	100%	1 [1,1]
frontier	median	>10	top 500	490	100%	2 [2,2]
frontier	min	3-10	top 500	8	100%	2 [2,2]
frontier	min	>10	top 500	490	100%	2 [2,2]
frontier	geo. mean	3-10	top 500	8	100%	1 [1,2]
frontier	geo. mean	>10	top 500	490	100%	3 [3,3]
+10 pts	mean	3-10	top 500	8	100%	3 [3,3]
+10 pts	mean	>10	top 500	490	21%	6 [6,6]
+10 pts	median	3-10	top 500	8	100%	1 [1,1]
+10 pts	median	>10	top 500	490	39%	3 [3,3]
+10 pts	min	3-10	top 500	8	75%	2 [2,2]
+10 pts	min	>10	top 500	490	70%	2 [2,2]
+10 pts	geo. mean	3-10	top 500	8	100%	3 [2,3]
+10 pts	geo. mean	>10	top 500	490	27%	5 [5,6]

Table 7: Evaluation-noise Monte Carlo (100 draws) and task bootstrap (OLLB 200, HELM 100 draws) on 10 targets at the +10-point cap: feasible share and median K^* [IQR] over draws against the unperturbed values for the same targets. Conditional sensitivity analyses on few targets, not validation of the cap assumption.

analysis	rule	draws	targets	feas.	K^* med [IQR]	feas. obs.	K^* obs.
noise HELM	mean	100	10	100%	18 [7,24]	100%	16
noise HELM	median	100	10	100%	6 [3,9]	100%	4
noise HELM	min	100	10	56%	2 [1,3]	70%	2
noise HELM	geo. mean	100	10	100%	18 [6,23]	100%	16
noise HELM	Borda	100	10	100%	12 [6,19]	100%	20
noise HELM	Condorcet	100	10	100%	13 [7,17]	100%	17
noise OLLB	mean	100	10	100%	3 [2,3]	100%	3
noise OLLB	median	100	10	100%	1 [1,1]	100%	1
noise OLLB	min	100	10	49%	1 [1,2]	70%	2
noise OLLB	geo. mean	100	10	99%	3 [1,4]	100%	3
noise OLLB	Borda	100	10	97%	1 [1,2]	100%	2
noise OLLB	Condorcet	100	10	95%	3 [3,4]	100%	3
taskboot OLLB	mean	200	10	87%	2 [1,4]	100%	3
taskboot OLLB	median	200	10	69%	1 [1,2]	100%	1
taskboot OLLB	min	200	10	53%	2 [1,3]	70%	2
taskboot OLLB	geo. mean	200	10	80%	2 [1,4]	100%	3
taskboot OLLB	Borda	200	10	86%	2 [1,3]	100%	2
taskboot OLLB	Condorcet	200	10	53%	3 [3,4]	100%	3
taskboot HELM	mean	100	10	100%	18 [6,24]	100%	16
taskboot HELM	median	100	10	100%	6 [3,9]	100%	4
taskboot HELM	min	100	10	70%	2 [1,4]	70%	2
taskboot HELM	geo. mean	100	10	100%	17 [6,23]	100%	16
taskboot HELM	Borda	100	10	99%	20 [10,28]	100%	20
taskboot HELM	Condorcet	100	10	100%	18 [10,22]	100%	17

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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   Justification: The abstract and introduction state the setting, the three investment measures, the feasibility-first findings and that conclusions are conditional on the cap and scale (Sections 1, 4, 5); investment is declared a score counterfactual, not a training cost.

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

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   Justification: Section 5 lists the score-counterfactual assumptions (no spillover, fixed rivals), caps as assumptions, $n = 6$ and unrelated OLLB scales, one HELM benchmark, snapshots, sampled cohorts, the margin ε and the conditional nature of the bootstrap and noise analyses.

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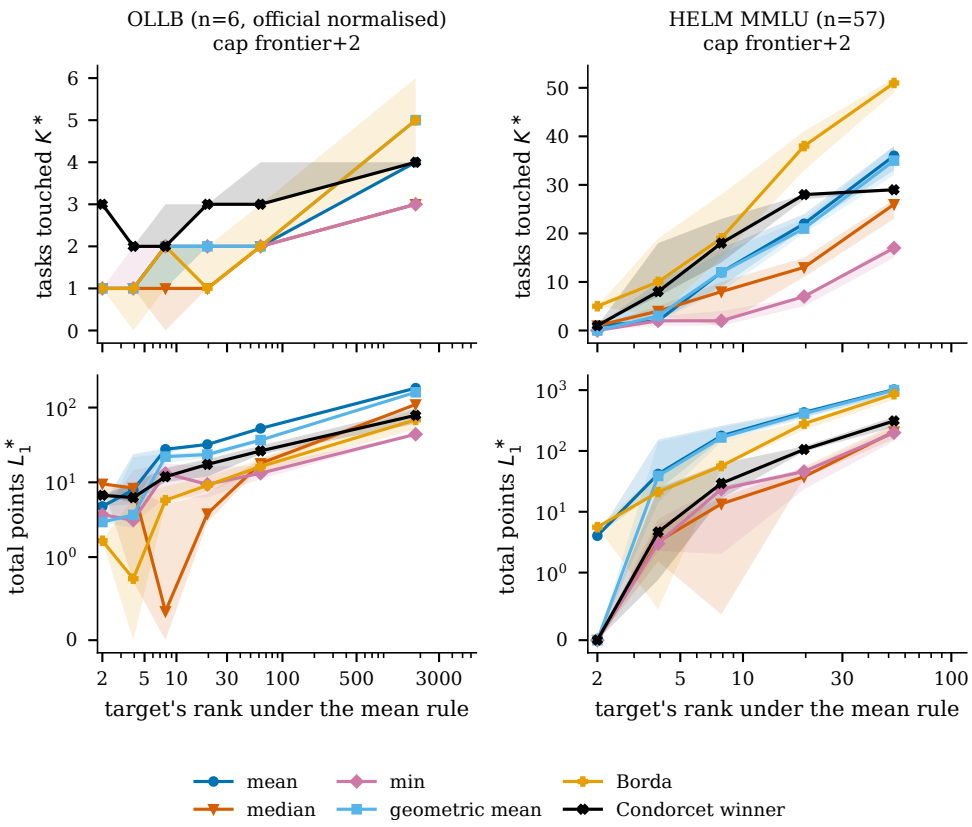


Figure 3: Minimum investment against the target's rank under the mean, cap frontier + 2 (every target feasible). Top: K^*; bottom: L_1^* (symlog axis; zero means already first). Lines: median over rank strata (2, 3–5, 6–10, 11–30, 31–100, >100), bands: 95% bootstrap interval. Left: OLLB (Borda and Condorcet on the 136-target sample, so their strata hold few targets); right: HELM MMLU. Cost rises with rank for every rule overall, not monotonically for the median and Borda L_1^* on OLLB at ranks 2–10, where intervals are wide.

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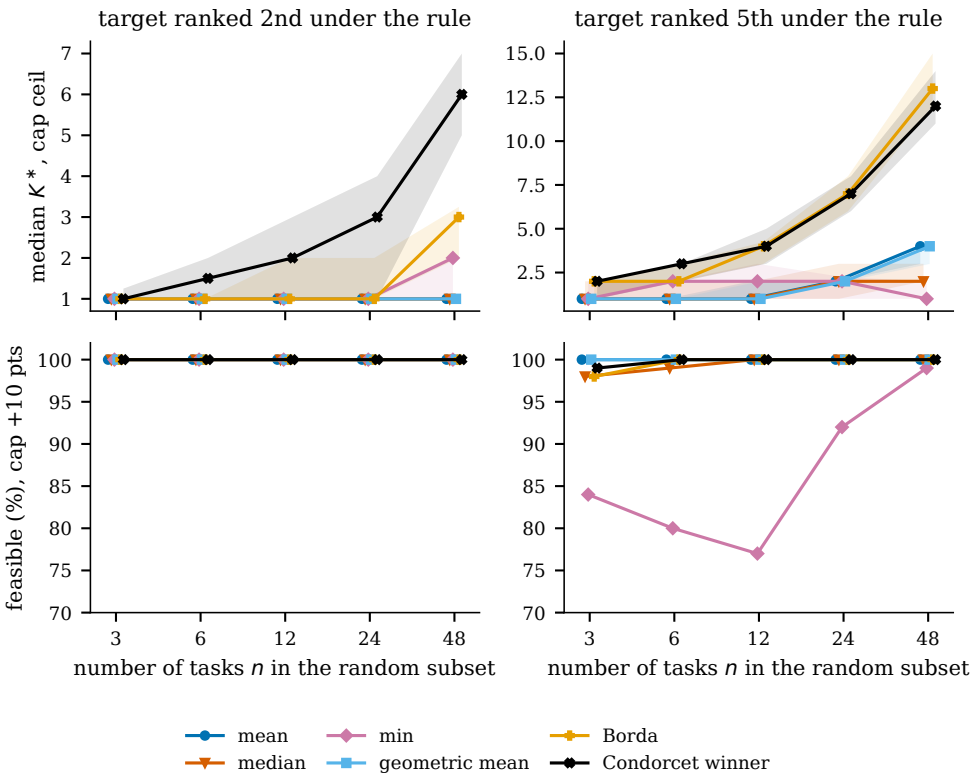


Figure 4: HELM MMLU with random subsets of n subjects (100 draws per n): for the model ranked 2nd (left) or 5th (right) under the rule itself in the subset, median K^* with IQR at the ceiling cap (top) and feasible share at the $+10$-point cap (bottom). One target per rule, draw and rank, so targets differ between rules. Borda and Condorcet need more tasks as n grows; for the 5th-ranked model the minimum rule is the only one whose feasible share falls clearly below 100%. Conditional sensitivity analysis over task subsets.

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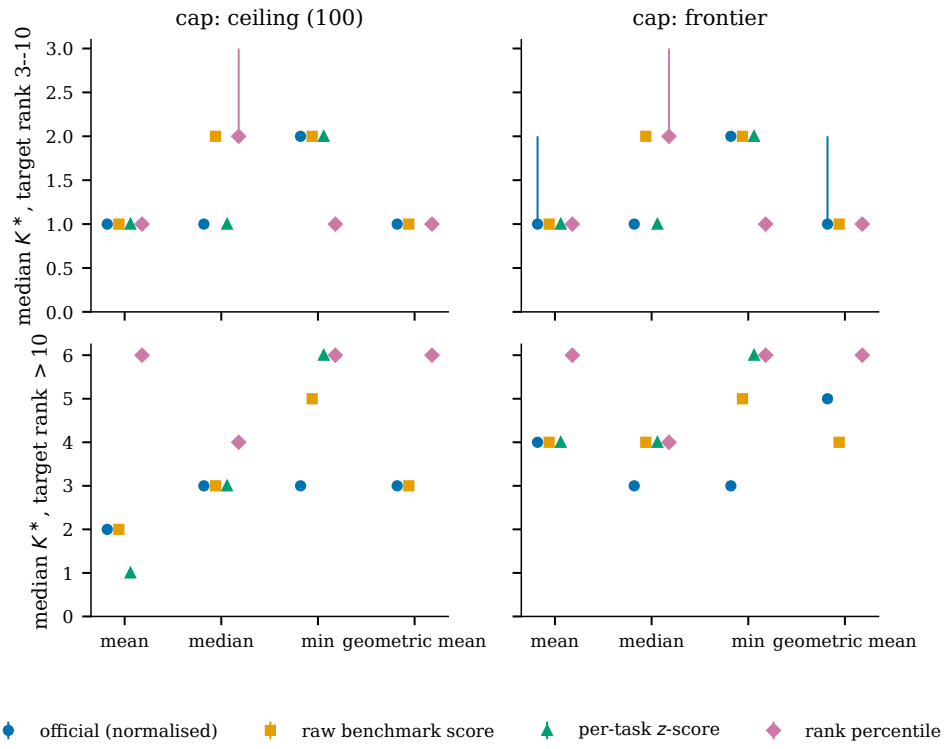


Figure 5: OLLB score rules on four versions of the task scores (official normalised, raw accuracy, per-task z-score, rank percentile): median K^* with 95% bootstrap interval, for targets ranked 3–10 (top; 8 targets) and >10 (bottom), ceiling cap (left) and frontier cap (right). All targets feasible; the geometric mean is not in the z-score set. Caps are in each variant's own units, so L_1^* is not compared across scales.

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